

Article 1 Summary

1)

Xie, H., Hwang, G. J. & Wong, T. L. (2021). Editorial Note: From Conventional AI to Modern AI in Education: Re-examining AI and Analytic Techniques for Teaching and Learning. *Educational Technology & Society*, 24 (3), 85–88.

2)

The question stated in the article is of the need for, and use of, AI in education.

3)

The article discusses the evolution of artificial intelligence (AI) in education, highlighting the shift from traditional rule-based models to modern deep learning techniques. Key findings include:

AI Transformation: Modern AI, particularly deep neural networks (DNN), has revolutionized both academic and industrial fields, leading to innovative applications in education (p. 1).

Integration Gaps: There is a knowledge gap between AI experts and educational researchers, making it challenging to integrate modern AI into educational practices (p. 2).

Research Trends: Despite the rapid increase in AI in education studies, few have focused on DNN applications, indicating untapped potential (p. 2).

4)

Keywords: Modern AI, AI transformation, Deep neural networks, Analytic techniques, AI in education

5)

Research into how AI affects pedagogical methods in teaching has not yet been conducted. There is a lack of overlap between researchers into pedagogical methods and researchers into modern AI. Because of this, very little is known about the best ways to teach in our modern world with AI.

This article is an editorial note for a journal that contains various articles on AI in education. Some of the various articles in the journal include data on "(i) AIED systematic review; (ii) modern AI applications; (iii) smart learning environments; (iv) AI-driven interventions; and (v) teaching/learning innovations for AI."

6)

I believe we can use this article to provide a general foundation for our recommendation report. It gives us various areas to focus on, in particular AI-driven interventions and smart learning environments. We believe that AI-driven interventions will prove particularly fruitful, as allow for personal – one-to-one – interactions and corrections. This has been already seen in many study tools, which use gen AI to create practice tests, in addition to providing feedback on student work.

Article 2 Summary

1)

Sarkar, S., Robin, S., Gokulrajan, R., Jitheshvar, M., & Jeshvan Kudiyaarasu. (2025). Generative AI for personalized learning experiences in education. In *2025 International Conference on Computing and Communication Technologies (ICCCT)* (pp. 1-15). IEEE.
<https://doi.org/10.1109/ICCCT63501.2025.11020445>

2)

The research addresses how generative AI (specifically Large Language Models and APIs) can be used to create personalized, adaptive learning experiences in higher education. The study investigates whether AI-generated learning materials in different narrative styles can improve student engagement, comprehension, and retention in software engineering education.

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- **Effectiveness:** The AI-powered personalization approach showed 10.2% improvement in predicting student feelings and 6.5% improvement in predicting student thoughts compared to baseline methods
- **Student Engagement:** Students who were more active with the AI-generated materials demonstrated higher engagement levels across all three content variants (academic, Batman-themed, and Wednesday Addams-themed)
- **Performance Metrics:** The model achieved F1-scores of 50704 for emotion prediction and 7349 for sentiment prediction on the testing dataset
- **Student Preferences:** Most students rated the personalized learning materials positively (scores of 4-5 on a 5-point scale), with particular interest in the character-based narratives
- **Accessibility:** The tool successfully integrated with existing Learning Management Systems (LMS), making it scalable and practical for implementation
- **Long-term Impact:** Follow-up surveys after 6 months showed students retained concepts better with the role-based teaching approach

4)

- Generative AI (GenAI)
- Personalized learning
- Large Language Models (LLMs)
- Natural Language Processing (NLP)
- Learning Management System (LMS)
- ChatGPT / GPT-4
- Adaptive learning
- Content personalization
- AI literacy
- Educational technology
- Finetuning
- BERT model

- Sentiment analysis
- Student assessment
- API integration
- Object-oriented programming (OOP)
- Constructivism
- Educational SDG4 (Sustainable Development Goals)

5)

- **Interesting Approach:** The use of pop culture characters (Batman, Wednesday Addams) to personalize educational content is creative and likely resonates with Generation Z students. This could be adapted to other contexts with different cultural references.
- **Ethical Considerations:** The paper mentions concerns about academic integrity, over-reliance on AI, and the digital divide. These are critical issues that need addressing before widespread implementation.
- **Methodology Concern:** The small sample size (only 20 students from 47 initially enrolled) limits generalizability. The authors acknowledge this limitation, but it raises questions about scalability.
- **Technical Implementation:** The integration with OpenAI's API and the three-variant content generation system is well-designed. The temperature setting of 0.7 balances creativity with accuracy effectively.
- **Teacher Role:** The study maintains the teacher as a quality-control checkpoint, reviewing all AI-generated content before student access. This hybrid approach seems practical and responsible.
- **Writing Quality:** The paper suffers from significant language issues and unclear phrasing (possibly due to translation or automated text generation), which ironically undermines credibility when discussing AI in education.

6)

This article is highly relevant for research on AI integration in educational settings, particularly for:

Connection to Personalized Learning Research: This study provides empirical evidence that generative AI can create engaging, personalized learning experiences without requiring individual teacher customization for each student. The three-variant approach demonstrates how AI can accommodate different learning preferences simultaneously.

Technical Implementation Framework: The methodology section offers a replicable framework for integrating generative AI into existing educational infrastructure through LMS systems. This practical approach is valuable for institutions considering AI adoption.

Student-Centered Design: The research highlights the importance of student agency in choosing their preferred learning style (selecting between narrative variants), which aligns with constructivist educational theories and modern pedagogical approaches.

Assessment and Metrics: The comprehensive metrics (engagement time, completion rates, sentiment analysis, retention after 6 months) provide a multi-dimensional framework for evaluating AI-assisted learning effectiveness beyond traditional test scores.

Limitations and Future Directions: Understanding the constraints (small sample size, single subject domain, limited to English, computational requirements) helps identify gaps that my research could address or avoid in methodology design.

This article would be particularly useful for developing arguments about the practical implementation challenges of AI in education and the importance of maintaining teacher oversight in AI-generated educational content.